What comes to mind when you think of precision medicine?

Nobody has responded yet.

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Precision Medicine & Health Justice

COLLEGE 117

April 78, 2026
Professor Daphne Martschenko
Professor Nicole Martinez

This week

Recognize key contextual factors likely to impact the development and implementation of equitable and just genomic and precision medicine

Apply the ethical frameworks covered in the course thus far to the ethical dilemmas raised by biotechnologies; identify and unpack the limitations of these frameworks in different global contexts

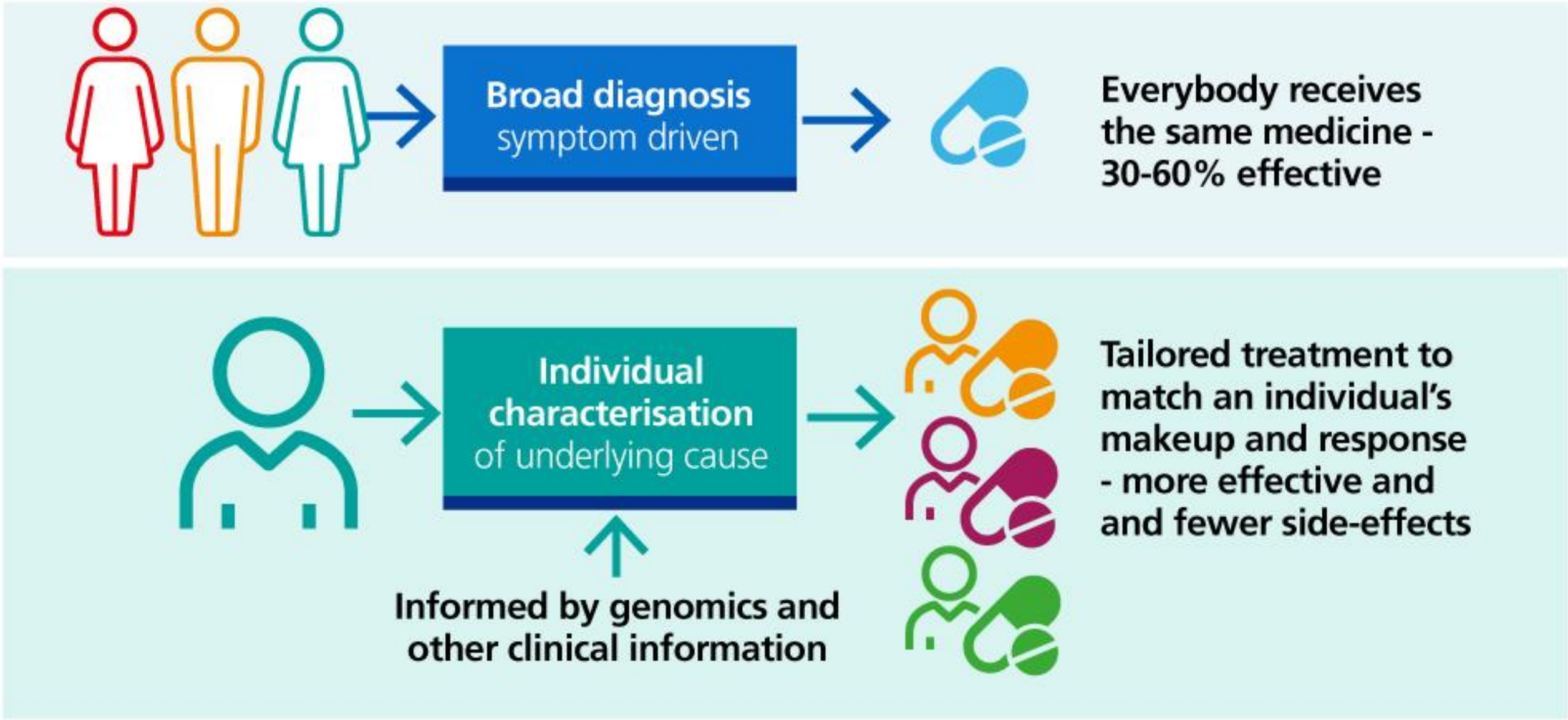
Today's guiding question

What factors impact the development and implementation of just genomic and precision medicine and in what ways?



Precision Medicine

"**Precision medicine** is rooted in the desire to improve health outcomes by producing and using multiple sources of personal data to better group and treat patients" (Bonham et al, 2016)



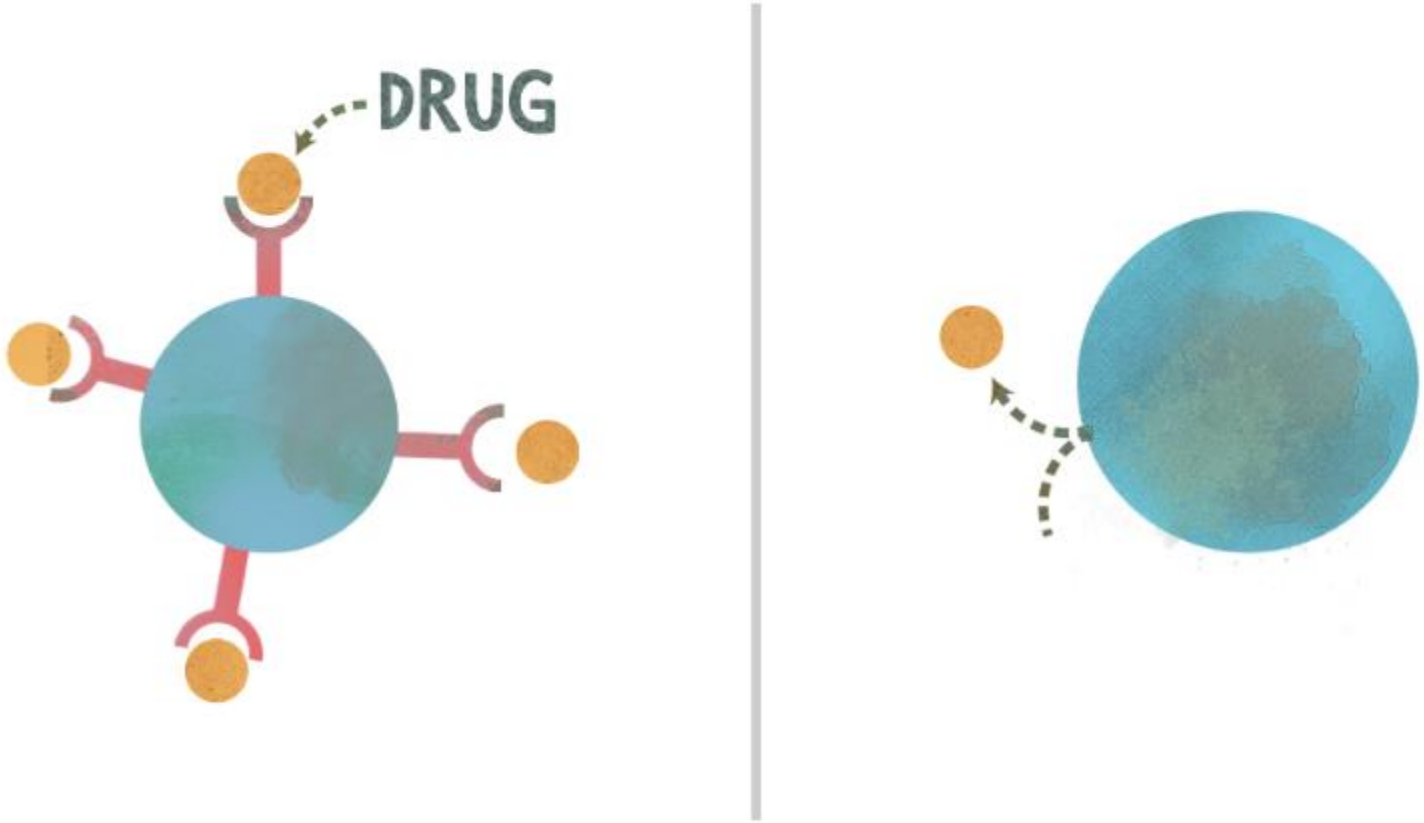
https://www.england.nhs.uk/long-read/accelerating-genomic-medicine-in-the-nhs/





Why Precision Medicine?

- There is variability in response to therapies
- More data will create better predictions about who will benefit
- Better resolution of diagnosis
- Use data to make more accurate, more precise assessment of patient and what's likely to help them

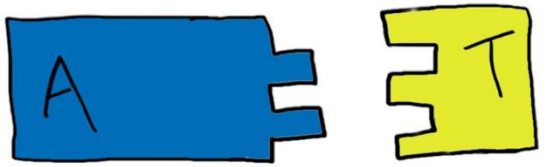


Targeted cancer drugs will bind to and kill cancer cells that have a specific surface marker (left) but not to cells that lack the surface marker (right). Because these drugs are concentrated near cancer cells, they produce fewer side effects than non-targeted drugs.

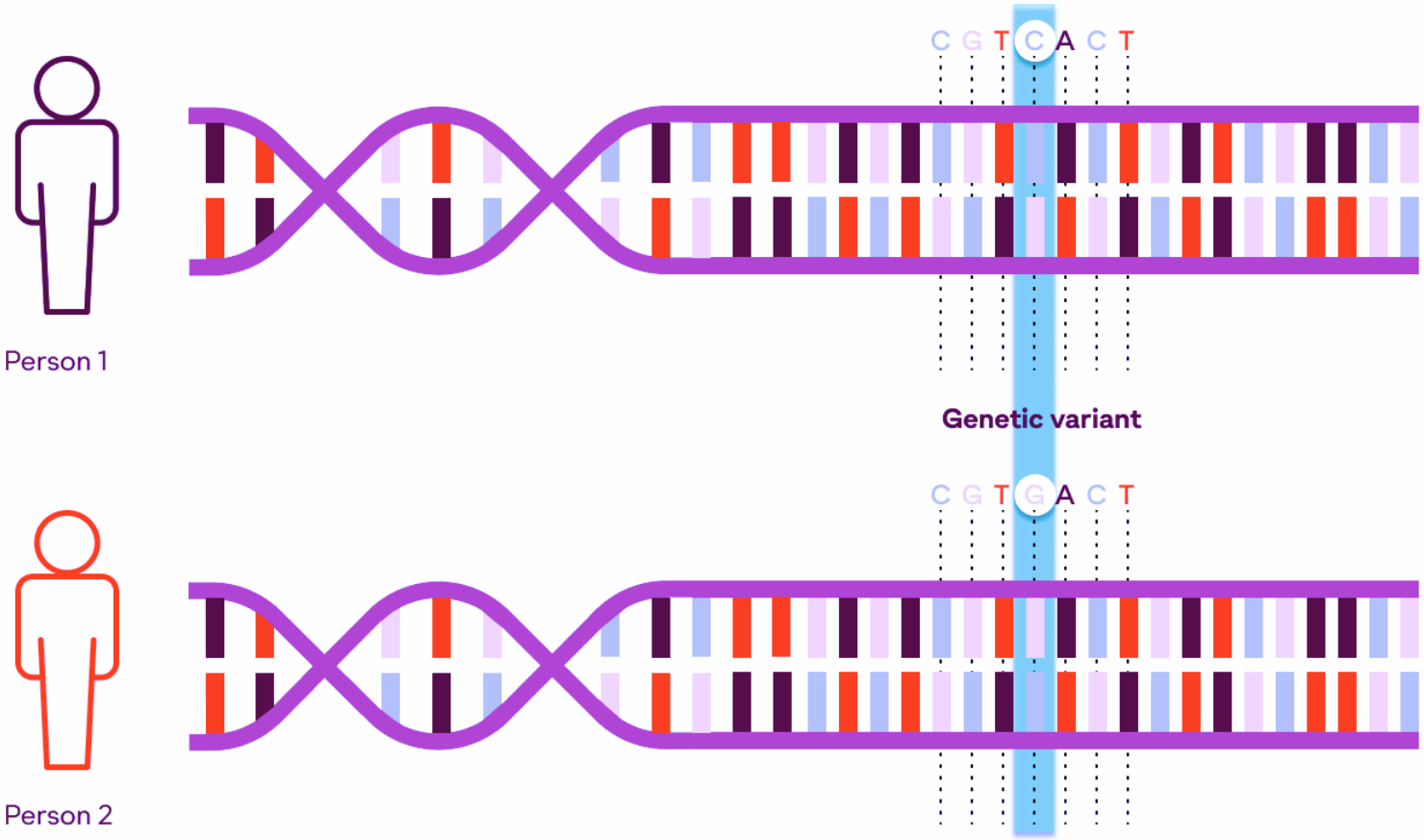
https://learn.genetics.utah.edu/content/precision/action/

Our genome is large! We are more genetically similar to each other than different

Our genome is made of 3 billion “base pairs”: places in the genome made of amino acids abbreviated A, T, C, and G



“a base pair”



- A single genome is like 1000 copies of the Complete Works of Shakespeare.
  - Printed out, that’s two tons!
- Comparing any two people on earth, they would have identical genomes at 99.8% of base pairs on average

Showing *that* genes help explain observed differences was important; discovering *which* genes is hard!

- **"Classical" twin studies** found that genes help explain observed differences (70s-80s)
  - For instance: Cancer, addiction, obesity

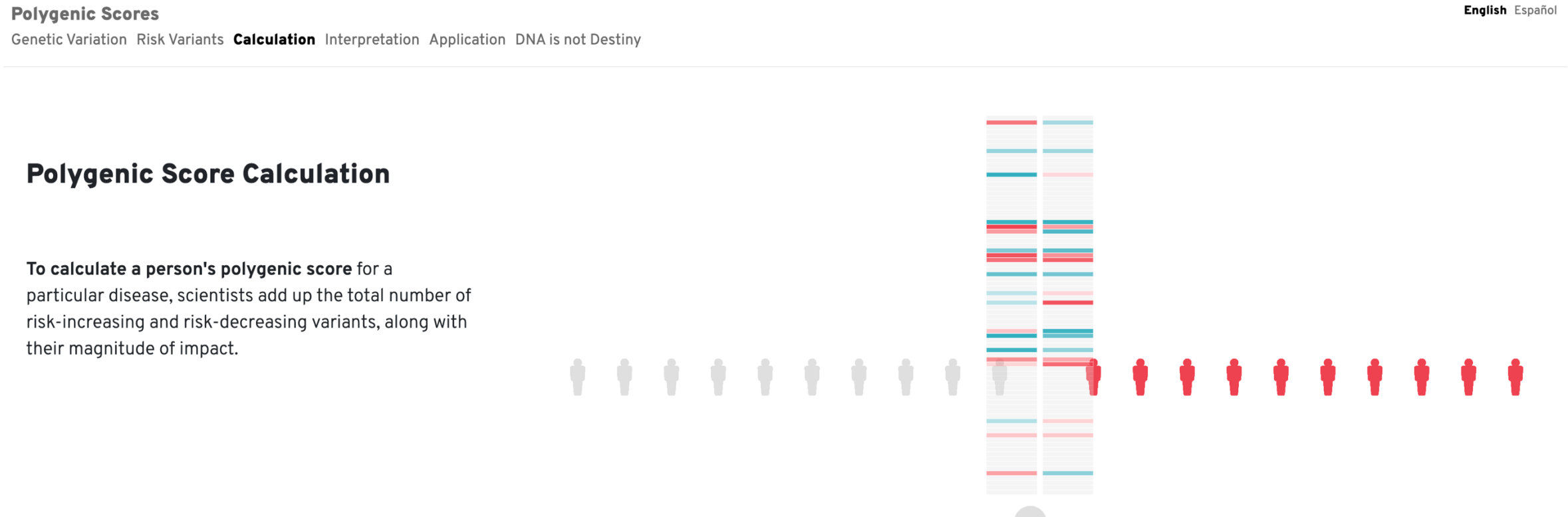
- 1st generation **molecular approaches** scored big wins (70s-90s)
  - single genes that explained rare diseases (Sickle Cell Disease, Cystic Fibrosis)
  - single genes with large effects on common diseases (BRCA and breast cancer)

- 2nd generation: **Genome-wide association studies** (GWAS) was disappointing (2000s)
  - Identified genetic variants that by themselves **explained** very little

- 3rd generation: **Polygenic scores** (PGS) not yet clear (2010-today)
  - May someday help explain observed differences

Polygenic scores "sum up" thousands of tiny effects of genetic variants to predict outcomes

Differences in *tens of thousands* or more genes each make a *very tiny* difference. When we add up all these tiny differences, they together can have a small, medium, or large influence on a health, social, or behavioral outcome



Polygenic scores depend on the environment. They are a snapshot of a particular moment in time & particular context

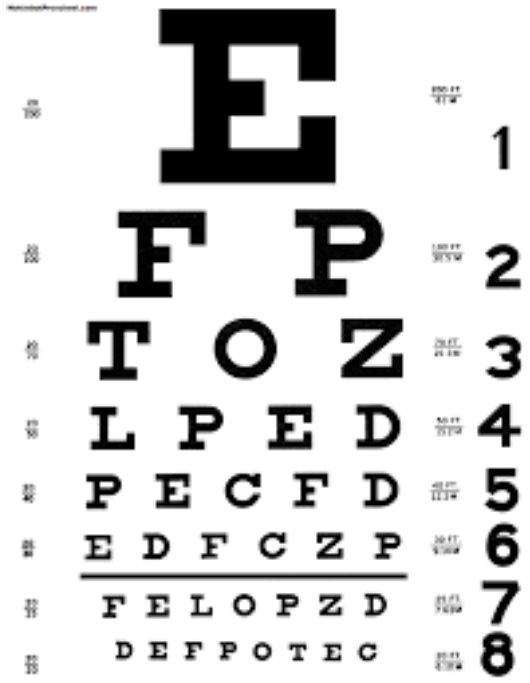
- A person might have a high PGS for type 2 diabetes when the society does not value exercise and where unhealthy foods are the more affordable and accessible option

- In a different world where regular exercise and healthy, unprocessed foods are commonplace, the exact same person might have a lower PGS for type 2 diabetes

- If the law requires or funds greater access to healthy foods or fitness programs, this will make a difference in a person's PGS too!

"Polygenic scores can and do reflect racism, sexism, or other prejudices, as well as more benign environmental factors" (Meyer et al, 2020)

Age



Genes







Trauma

Stress

Diet

Exercise

Precision medicine promises individualized medicine, but still relies on categorizing people according to ‘groups’ or ‘types’

Genetic Ancestry
Biological inheritance of DNA, can be traced through the genome using DNA data

Genetic Similarity
Similarity of one’s DNA, to some reference population(s)

Geographic Ancestry
Region(s) of the world where portions of an individual’s ancestors lived. As a result of migration of individuals and populations over time, an individual’s genetic ancestry might not match their geographic ancestry.

Genetic ancestry =/= Race

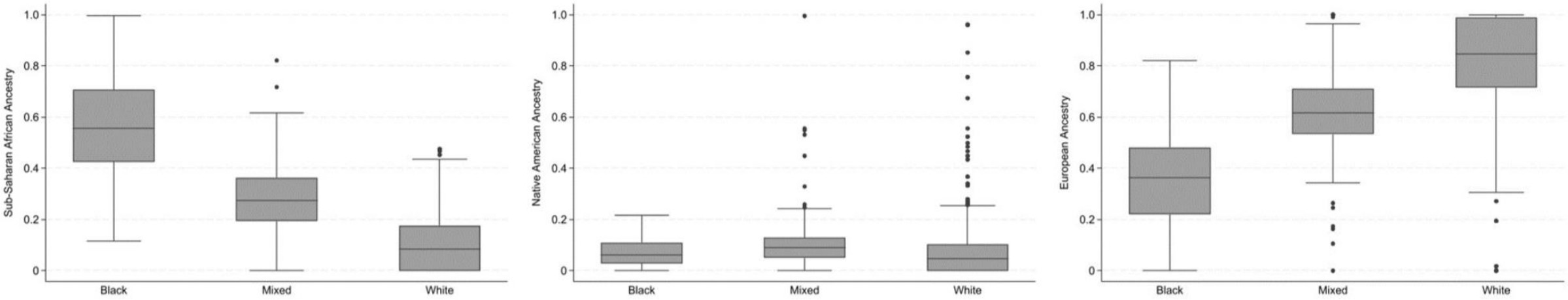
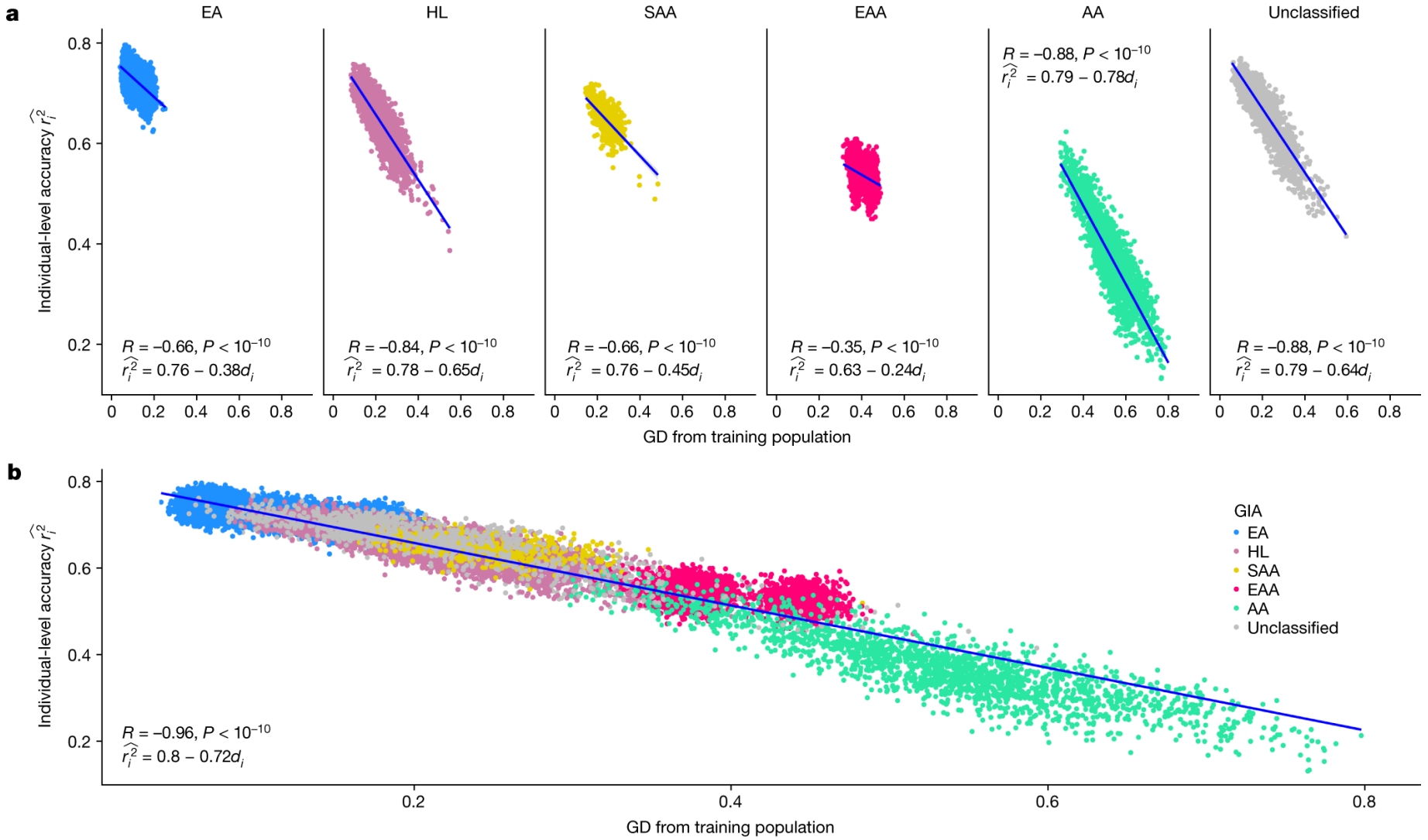


Figure 1. Box plot of median individual genetic Sub-Saharan African, European, and Native American ancestries according to self-reported skin color/race. The proportion of East Asian ancestry and the self-reported Indigenous, Yellow, and Not Answered categories are not presented due to the low frequency of observations. Detailed parameters (median, interquartile ranges, and *p*-values for statistical differences) are presented in Table 1.

Black Brazilians range from 0% European ancestry to over 80% European ancestry; White Brazilians range from 0% European ancestry to 100% European ancestry

Pereira et al (2024)

Differences **within** genetic ancestry groups



Ding Y, Hou K, Xu Z, et al. Polygenic scoring accuracy varies across the genetic ancestry continuum. *Nature*. 2023;618(7966):774-781. doi:10.1038/s41586-023-06079-4

NHGRI 2020 Strategic Vision

"Genomics will no longer use social constructs, such as race in human research studies"

- Bold Prediction #4



How might precision medicine enhance justice?

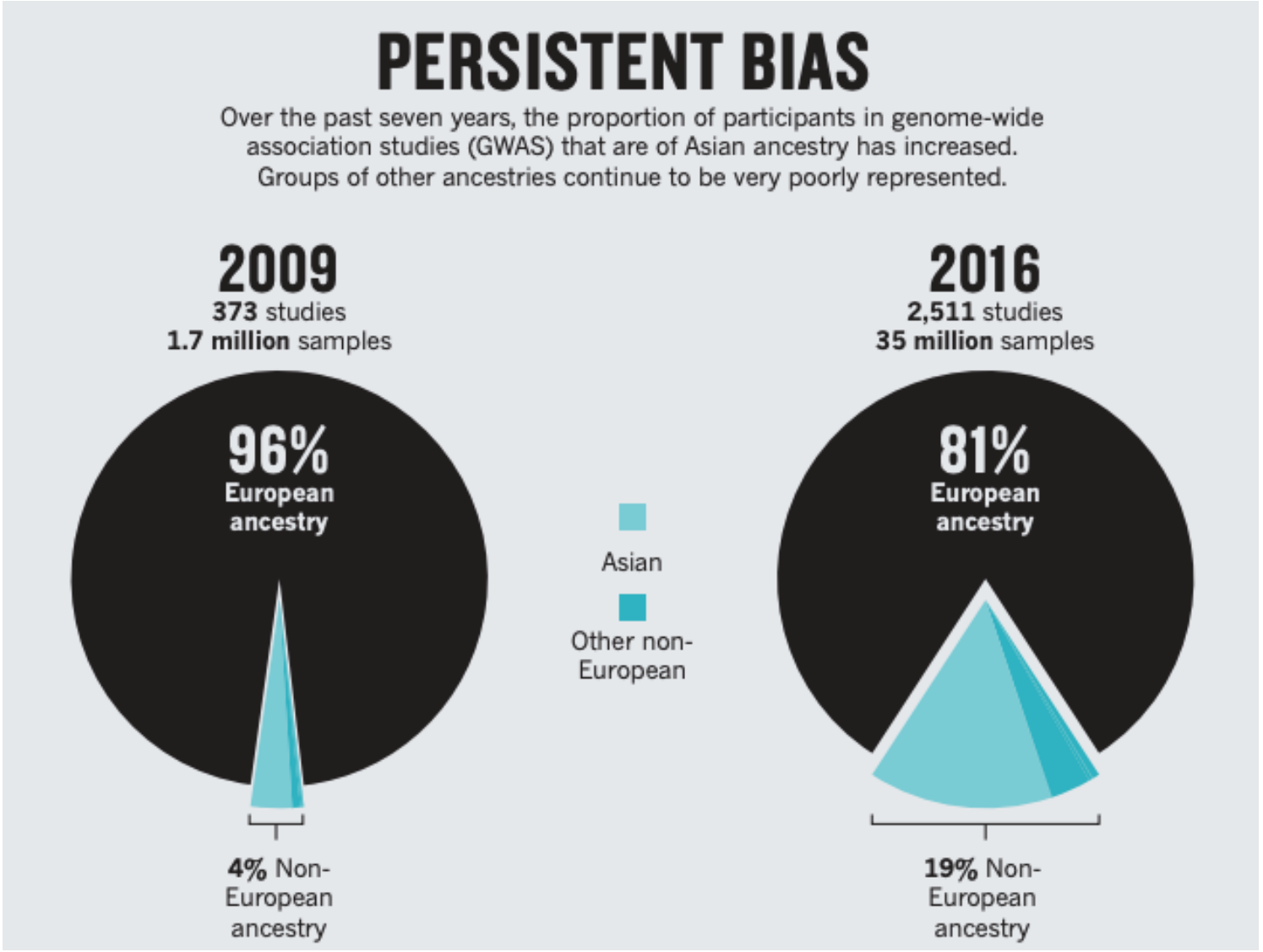
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Barriers to Justice in Precision Medicine

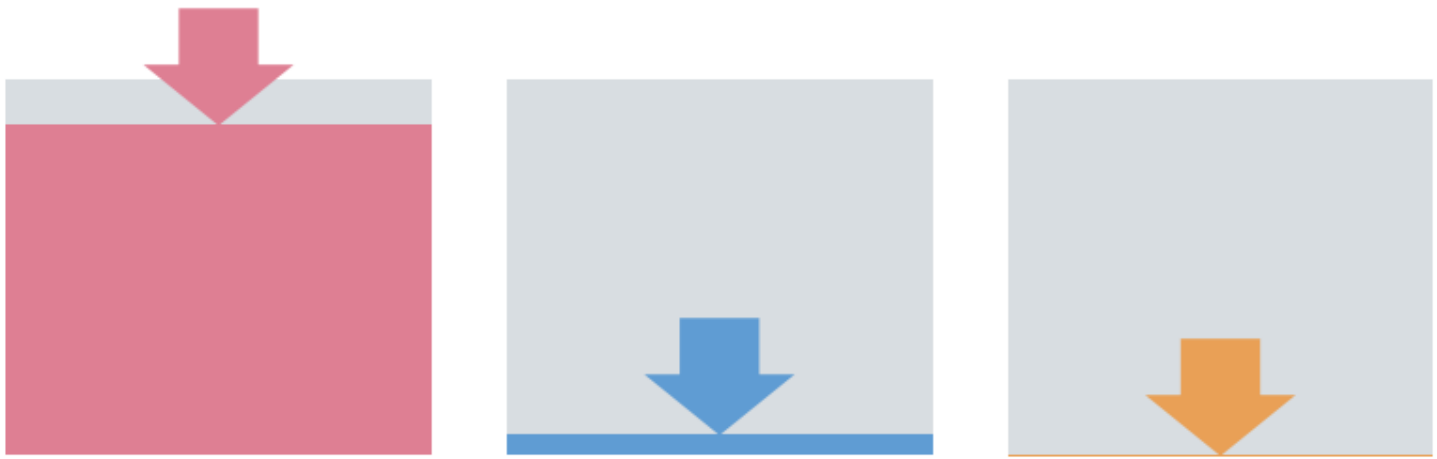
Bias in genomic data



GWAS
Diversity Monitor

Total GWAS participants diversity

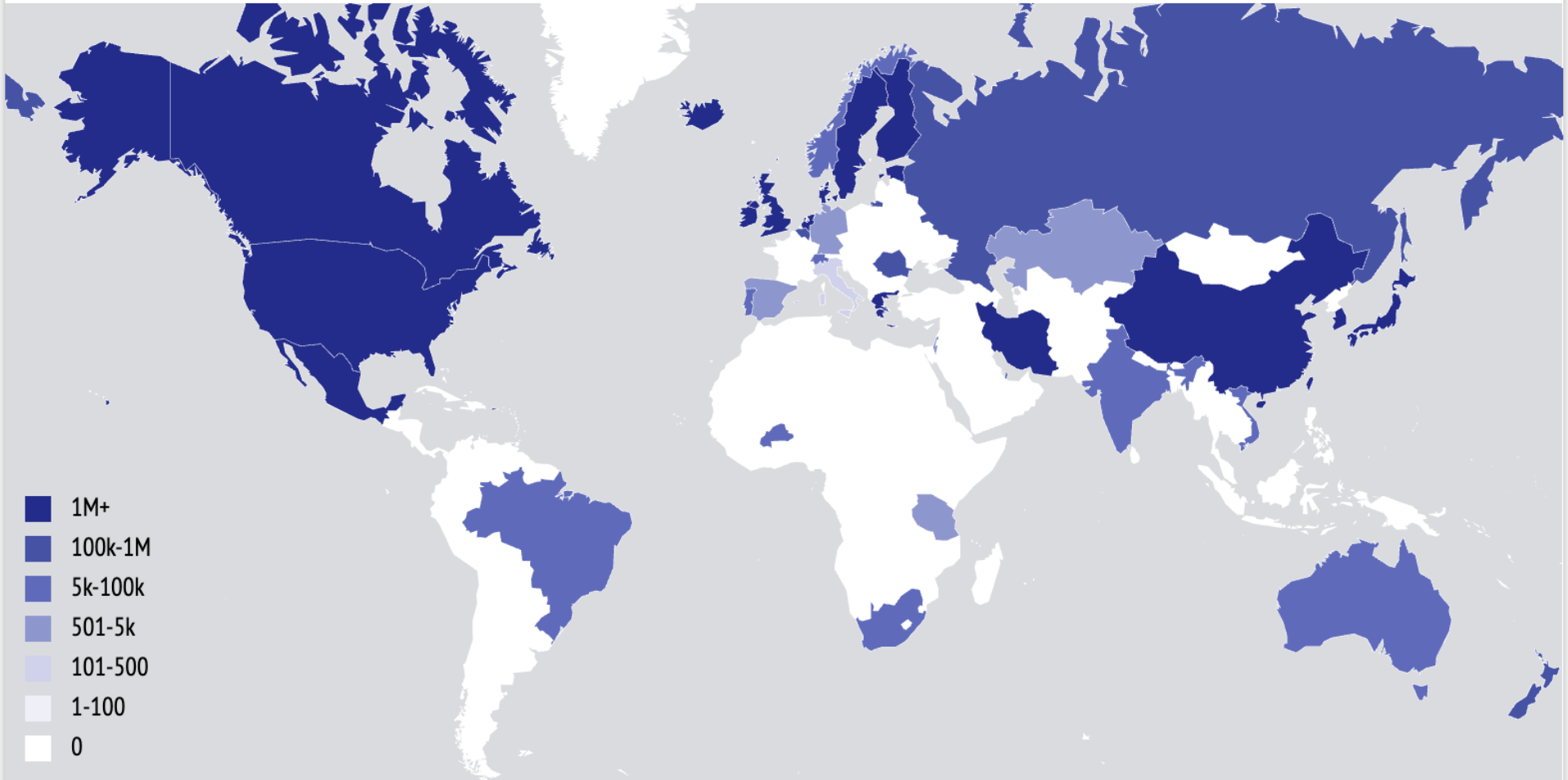
Version 1.0.0. Last check for data: 2025-09-05 00:34:26 .



87.77%	5.33%	0.16%
European	Asian	African

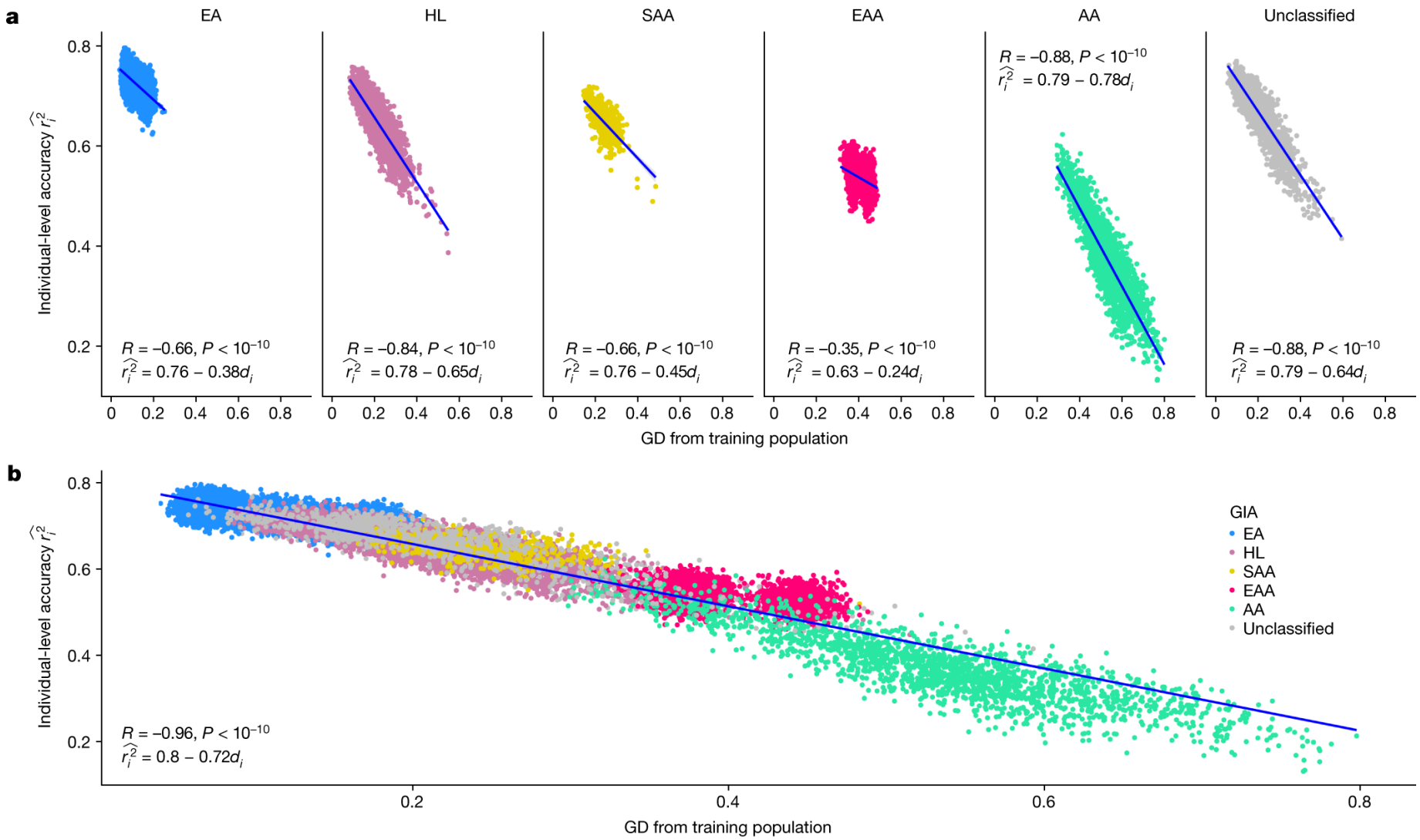
Participants by country (all parent terms summed across studies)

Both Stages



Please cite this as: Mills, M.C and Rahal, C., (2020). 'The GWAS Diversity Monitor Tracks diversity by disease in real time'. Nature Genetics, 52, 242-243. doi: 10.1038/s41588-020-0580-y

People and information lost when we discretize



The "unclassified" are **left out**

Ding Y, Hou K, Xu Z, et al. Polygenic scoring accuracy varies across the genetic ancestry continuum. *Nature*. 2023;618(7966):774-781. doi:10.1038/s41586-023-06079-4



- One of most widely used resources for biomedical research
- A volunteer-based study



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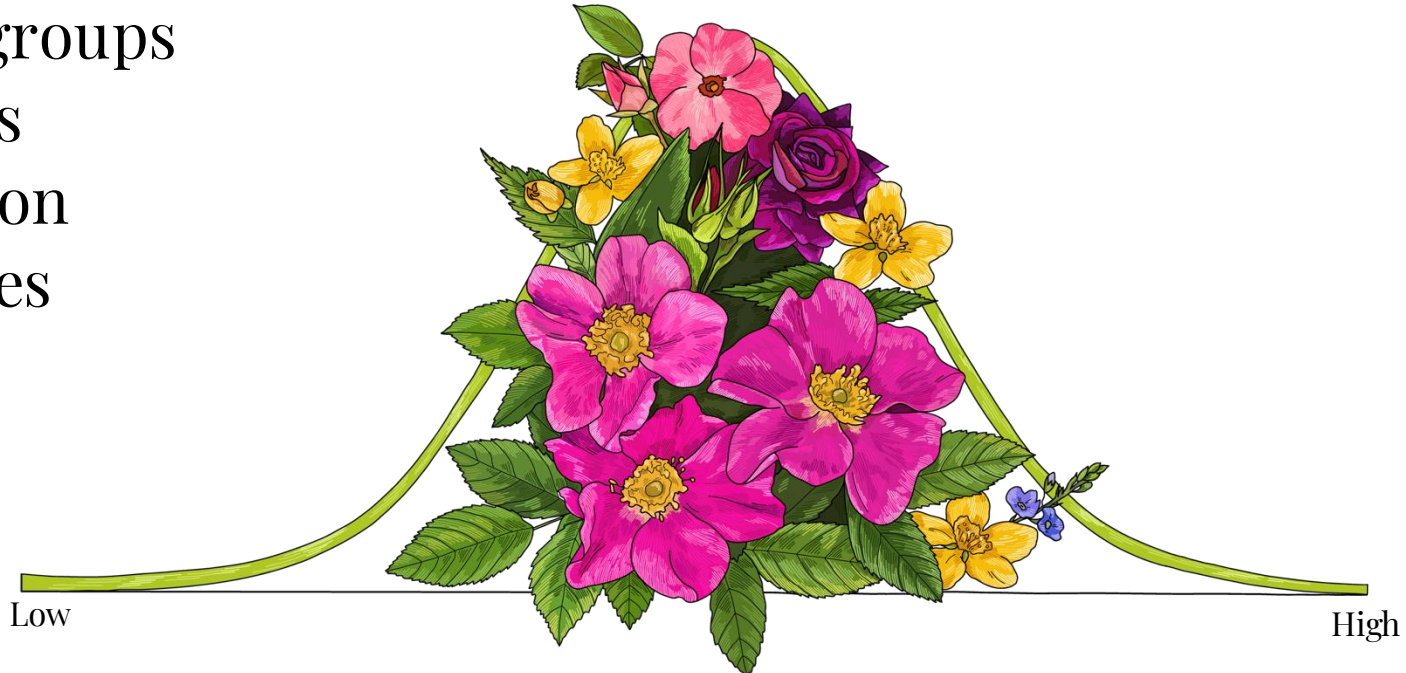
Limitations:

- Study participants are not representative of the target population (called ‘volunteer bias’ or ‘selection bias’)
  - Of the 9 million invited to take part, only 5.5% did
  - Typically healthier, wealthier, more educated than the target population
  - Overrepresentation of women and older age groups
  - Underrepresentation of people from rural communities
  - Almost exclusively European ancestries

A technology like a polygenic score won't be as accurate for:

Groups that these scores haven't been developed on, such as:

- Different age groups
- Different environmental contexts (i.e., urban vs. rural)
- Different socioeconomic groups
- Different gender identities
- Different levels of education
- Different genetic ancestries



Data biases are driven by...

History

- Instances of research abuse and exploitation have cultivated mistrust: impacts researchers' abilities to work with diverse communities (think of 'Geneticist's Dilemma article)

- Limited understanding of genetic concepts & limited data on effective models for community engagement



Data biases are driven by...

Resource Allocation Issues

- Large-scale genetic studies are expensive and time intensive
- Lack of funding, institutional capacity, and a skilled workforce in LMICs

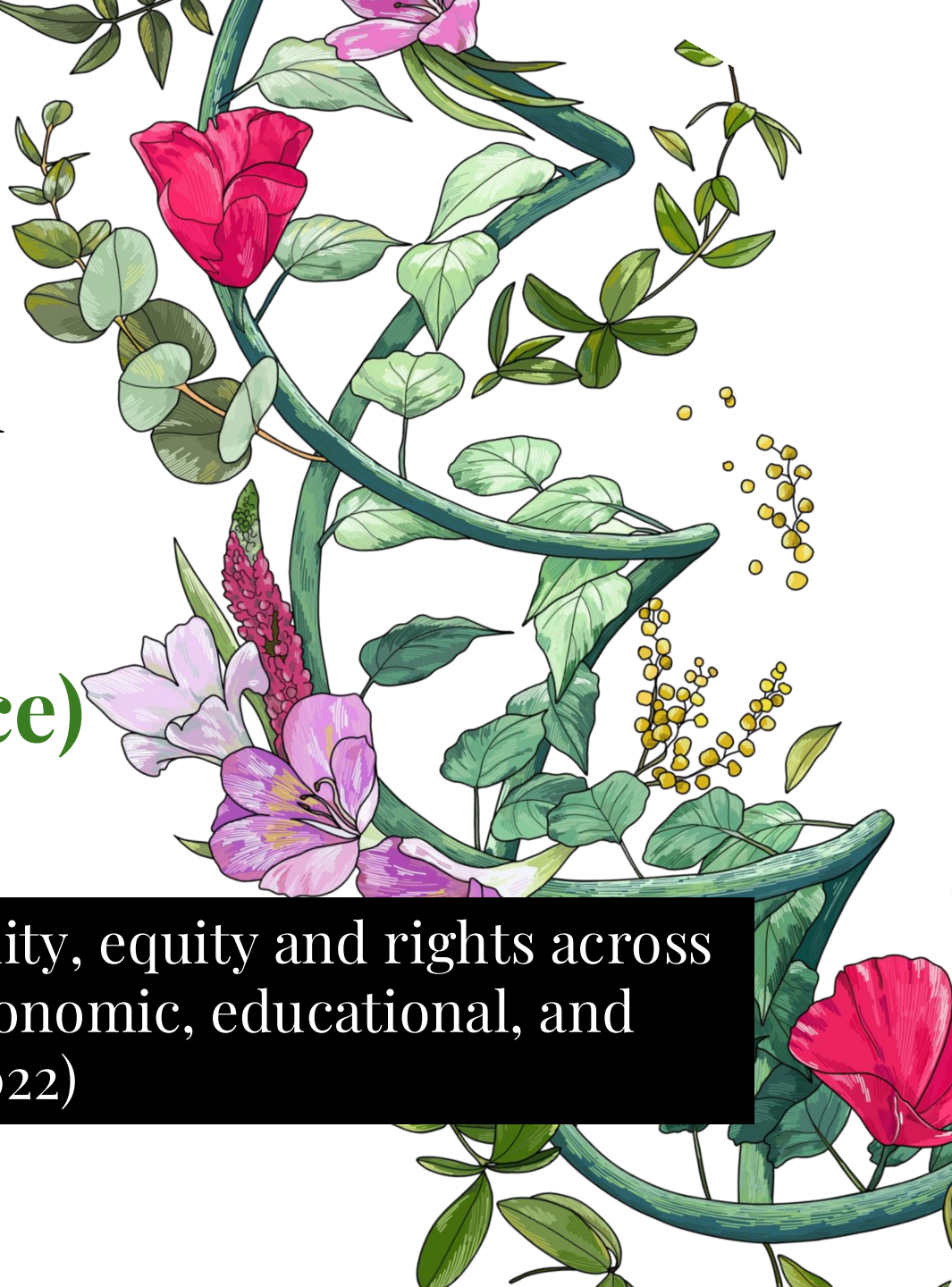


Biases will...

- Fail to make the benefits of genomics accessible (Popejoy & Fullerton, 2016; Fatumo et al, 2022)
- Risk exacerbating health disparities if translated into clinical settings (Martin et. al, 2019)
- Result in major missed scientific opportunities (Fatumo et al, 2022)



Disparities in data collection (research injustice) produce disparities in real-world implementation (social injustice)



Social justice: promoting fairness, equality, equity and rights across multiple aspects of society, including economic, educational, and workforce opportunities. (Killen et al, 2022)

For instance: Genetic test availability & performance

- Some genetic tests limited to groups of European ancestries
- Higher rate of ‘uncertain’ results in non-European ancestry populations



Even if research justice is secured, social justice issues may remain



For instance: Access to genetic services & treatments

- Racial and ethnic minority groups, women, people living in rural communities, people who are uninsured or underinsured, and those with lower education and income are less likely to access genetic services (Khoury et al, 2022)
- Sickle Cell Gene Therapies & cost

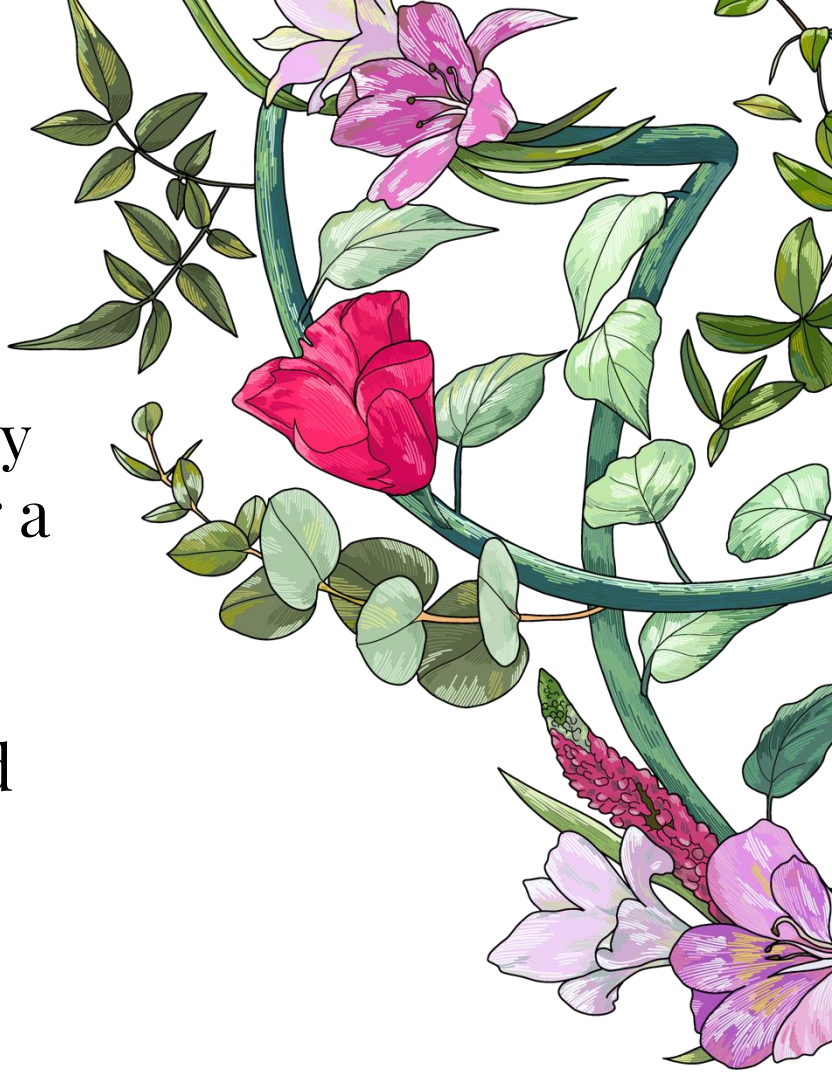


Efforts to Enhance Justice in Precision Medicine

On justice

- What strategies are being employed to improve access to technologies that people are systematically deprived of such that the are prohibited from living a decent life (think Faden & Powers)

- What strategies are being employed to promote and protect health in a population and thereby create opportunities (think Daniels)



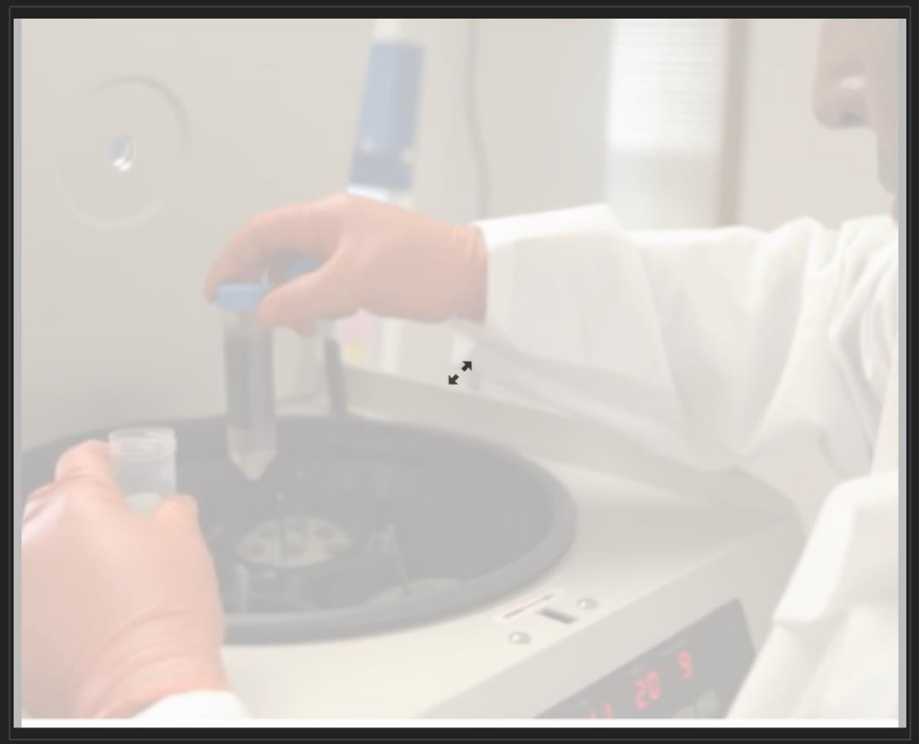
A sufficiently decent life = a life that is not subjugated, not marginalized, not stigmatized or where one is deprived of full use of their faculties

Native BioData consortium

RESEARCH FOR NATIVES, BY NATIVES

centrifugeonepager

The **Native BioData Consortium (NBDC)** is the first 501(c)(3) nonprofit research institute led by Indigenous scientists and tribal members in the United States. As a biorepository (or "biobank"), we ensure that advances in genetics and health research benefit all Indigenous people.



INDIGENOUS-LED GENOMICS

Native Americans suffer worse health outcomes than any other group

…particularly in cancer, heart disease, and autoimmune disorders. As science continues to innovate, this gap in health will continue to widen unless we intervene.

- **Pharmacogenomics** (the genetic study of drug efficacy) –Our research can find unique genetic signatures that contribute to chronic diseases and improve drug treatments.
- **COVID-19 and public health –** We can help Native communities become better-equipped to fight emerging infectious disease and improve public health interventions.
- **Create an education center** to train Indigenous students, who will become scientific leaders for our people in technology, agriculture, and health.

> The enormous "rescue" and promise of technologies to improve health care crises in Native communities is here but will not be realized unless tribes take control of their data.



Native BioData
consortium



$176million
Overall monetary investment by NIH/Wellcome Trust

+500
Consortium Members

310
Study Coordinators/Community Engagement Representatives/Fellows Travel supported

201
Workshops/Themed meetings (H3Africa Only)

51
H3Africa Projects supported

1026
Number of Workshop/Satellite meeting attendees - H3Africa Only

260
Number of BSc

1528
Number of members registered on the HtrainDB

467
Trainees

118065
Research Participants Recruited



H3 Africa & NativeBio Data

Resources devoted to infrastructure & capacity building

- More data collection can translate to **more rigorous & representative research, research justice, and technological/scientific innovation**

- Focus on infrastructure and capacity building within the affected community also promotes **social justice**
  - Examples of **inclusive design** AND **co-design**. Latter can identify what is needed to translate research into practice (i.e., what is needed to achieve social justice and fairly distribute benefits)

Other examples

Academic Publishing & Funders

- Funders and high-profile journals seeing GWAS done on populations that are overrepresented in genomic databases as less innovative or novel
- Rewarding development of genomic analysis methods that don't rely on discrete categorization of genetic ancestry

All of US Research program in US

- Effort to get samples from up to 1 million individuals that are representative of US population
  - E.g., efforts: mobile vans in rural communities

Analog: NIH Revitalization Act of 1993

- Federal law requiring that women (and minorities) be included in all NIH-sponsored clinical research.

- NIH would not fund any grant, cooperative agreement, or contract or support any intramural project unless it complied with this policy.

Analog: Navajo Moratorium on Genetics Research 2002

- The Navajo Nation placed a moratorium on genetic research studies in 2002
- Covers genetic investigations, research, or studies conducted within the Navajo Nation, stipulating that no such research can occur until a comprehensive Navajo Nation Health Research Code is amended and approved.
- February 2018, the Navajo Nation Council established a Genetics Policy Working Group

In summary:



Disparities in data collection produce disparities in real-world implementation

- There are many factors contributing to data collection disparities (social/contextual factors, capacity-building, infrastructure & access)
- **Native BioData** & **H3 Africa** are efforts to address disparities in data collection & examples of **inclusive designs**

Even if data quality is improved, real-world implementation issues may remain

- Access, funding, capacity & infrastructure issues
- **Native BioData** & **H3 Africa** are also examples of **co-designs** that seek to address social justice issues

In summary:

- Research (in)justice and social (in)justice shaped by history/social context, resource allocation, and infrastructure issues

- **Social Justice:** May persist even when justice in research is improved



What's a question you have? Upvote entries that you think are most important

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